

L7 - ML - Regularization

Date:

2/10/24

Cross Validation

Recap

- Gradient Descent variants
- Polynomial Regression
- Generalization & Occam's Razor
- Underfitting & Overfitting
- Bias-Variance Tradeoff

→ In this lecture

Agenda

- Regularization
- hyperparameter tuning
- cross validation

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(I)

⇒ overfit

Training Error - ~~low~~ low
Testing Error - high

model variance (change) → high
→ Not understood this

(II)

⇒ Underfit

Training Error - ~~high~~ high
Testing Error - high

model variance (change) → low

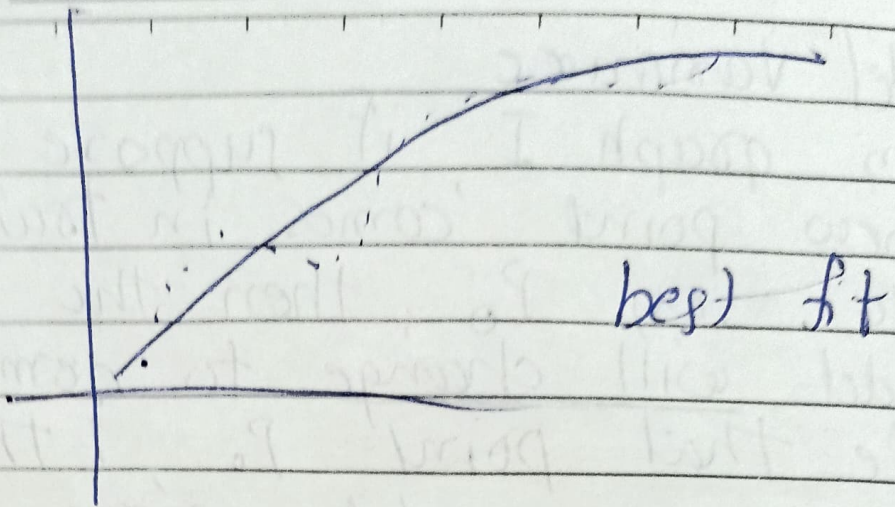
Model Variance

In graph I, if suppose a new point 'comes in training data say P_0 , then the model will change to accommodate that point P_0 . I think change in model mean change in weights.

In graph II, if suppose a new point 'comes in training data say P_0 , then the model will not change to accommodate that point P_0 .

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III



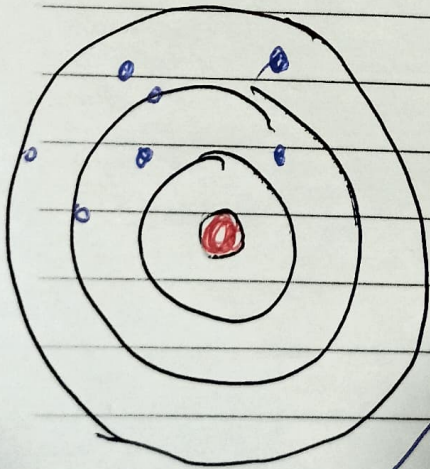
Training error - low
testing error - low

Model variance :- low

Bias & Variance

Case I

relates to
↓
training error



High bias

high loss since target
is missing everytime

High Variance

high variance since
hitting is not consistent
seems random

how model
predicts
varies each time

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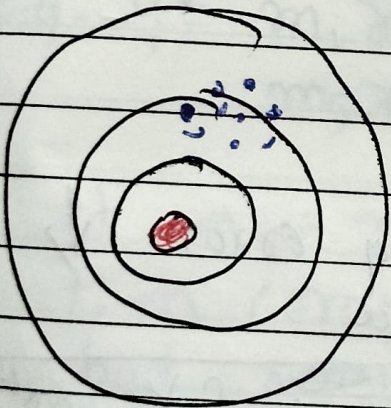
case II



Low bias

high variance

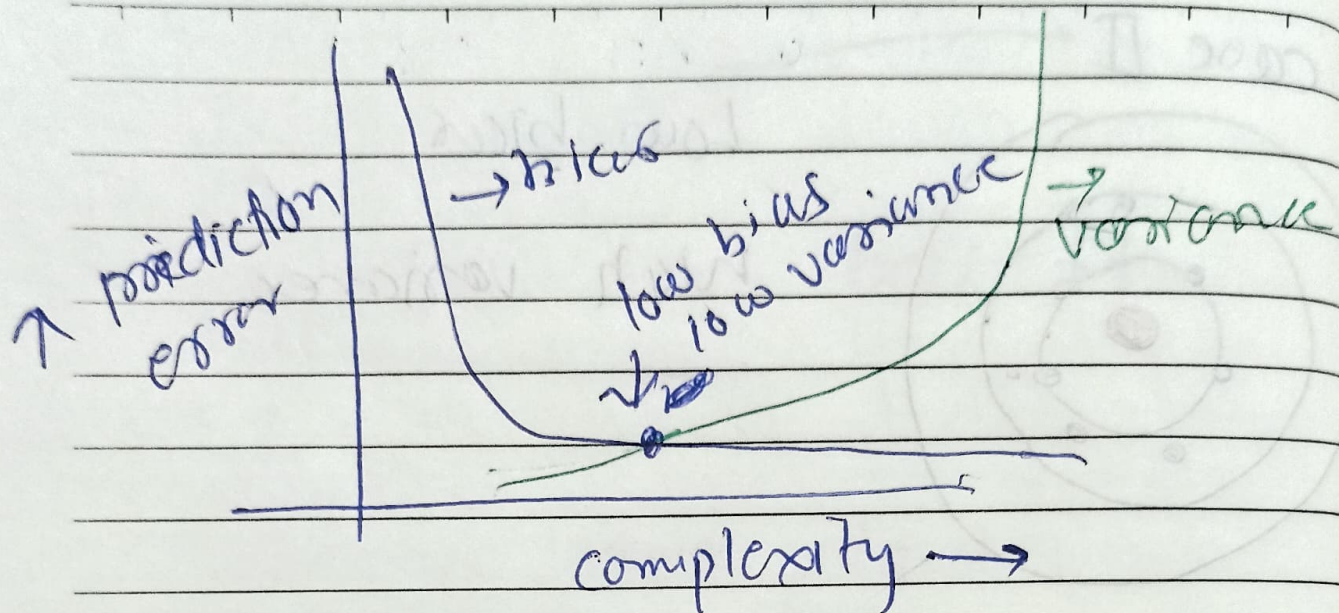
case III



high bias

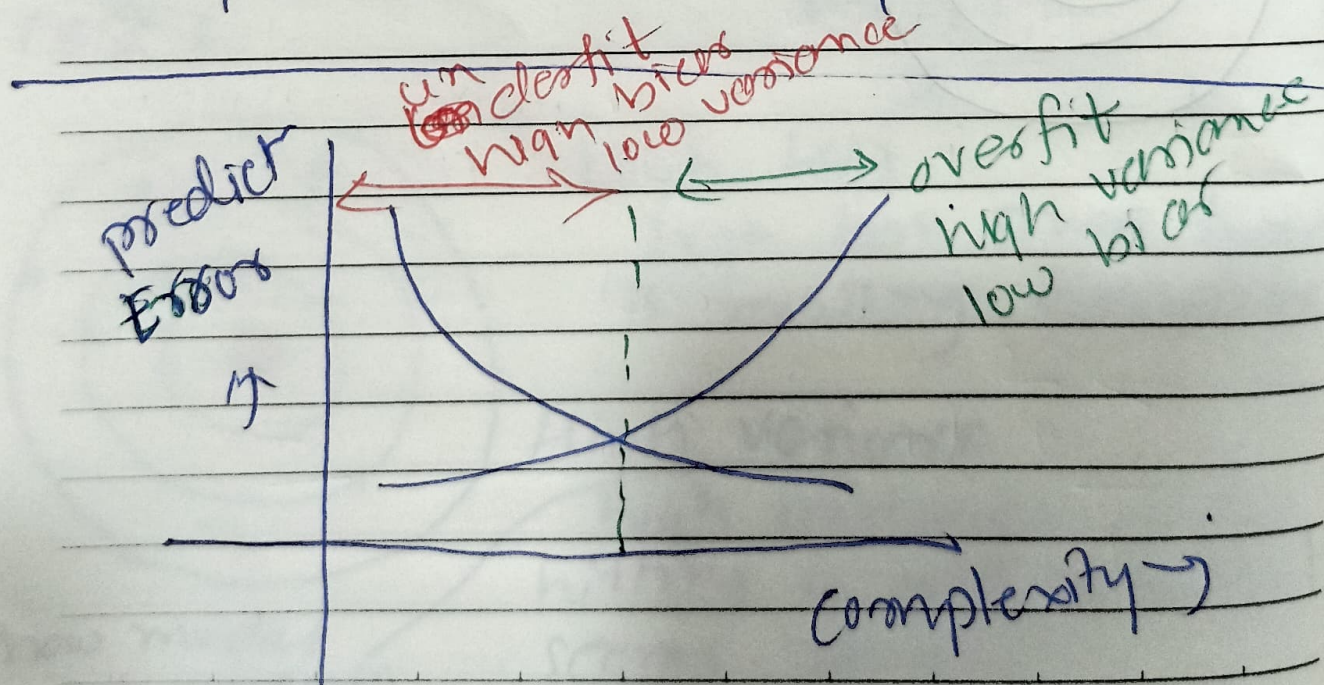
low variance

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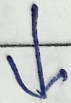
complexity $\rightarrow$ more parameters,
high degree of parameter
overfit

complexity $\uparrow$ training error $\downarrow$

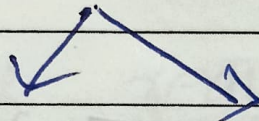


Over fit to best fit

high variance low bias
major issue not the issue



caused by complex model



more features more degree/order

$$\text{e.g. } y = m_1 x_1^2 + m_2 x_2^2 + m_3 x_3^2 + m_4 x_4^4 + m_5 x_5 + m_6 x_6 + m_7 x_7$$

if $m_4 = m_5 = m_6 = m_7 = 0$

then complexity ↓

but ~~rem~~ it removes important concept so make it very small.

$$m_4 = m_5 = m_6 = m_7 = 0.001$$

this will help to remove complexity.

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How to do it. & which parameter to do it, ?

$$\text{loss} = \text{MSE}$$

$$= \frac{\sum (y - \hat{y})^2}{N}$$

$$= \frac{\sum (y - w^T x)^2}{N}$$

$$= \frac{1}{N} \sum (y - w_1 x_1 + w_2 x_2 + \dots)^2$$

the problem of overfit is weight values we want weight to be lower to avoid overfit so to do that we will add penalty to loss function

$$= \frac{1}{N} \sum (y - w_1 x_1 + w_2 x_2 + \dots)^2 + \sum_i w_i^2$$

Here if model tries to $\uparrow$ weight
the loss $\uparrow$ & it will not be
good model

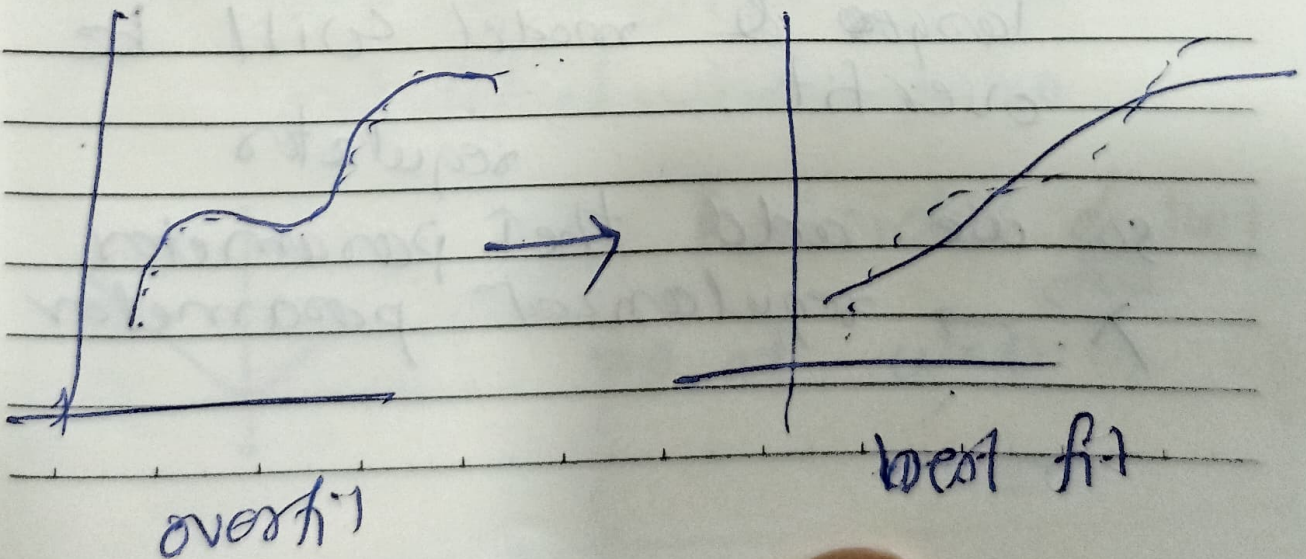
So ~~The~~ model learn weight
in this way which will less &
avoid overfit.

This is call regularization.

$\downarrow$
overfit to best fit

By adding penalty $\sum w_i^2$, does
the model accuracy affect?

$\rightarrow$ Yes. but we get smooth
graph, which more important
Here bias (error) will increase



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New loss function

$$\text{loss} = \text{MSE} + \text{Regularizer}$$

$$= \frac{\sum_i (y - \hat{y})^2}{N} + \sum_i w_i^2$$

if $\sum_i w_i^2$ is very high

all the weights will be close to 0, and model will be underfit.

if $\sum_i w_i^2$ is very low

all the weight will be large & model will be overfit.

regularizer

so we add the parameter

$\lambda \rightarrow$ regularization parameter

is λ high ~~overfit~~ underfit
 λ is low ~~underfit~~ overfit

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$\lambda \uparrow$ $w \downarrow$ underfit

$\lambda \downarrow$ $w \uparrow$ overfit

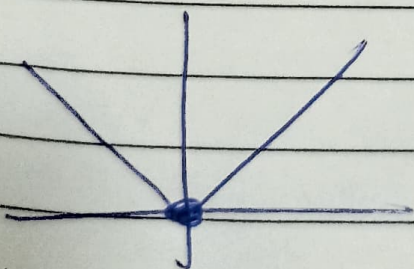
$$\text{loss} = \text{MSE} + \lambda \sum w^2 \quad \left. \begin{array}{l} \text{L2} \\ \text{Regularization} \end{array} \right\}$$

$$\text{loss} = \text{MSE} + \lambda \sum |w| \quad \left. \begin{array}{l} \text{L1} \\ \text{Regularization} \end{array} \right\}$$

$$\frac{d}{dw} |w| = 1 \quad \text{when } w > 0$$

$$= -1 \quad \text{when } w < 0$$

not defined when $w = 0$



so consider $\frac{d}{dw} |w|$

$= 0$ at $w = 0$

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here weight can be 0

$$\text{If } y = w_1 x_1 + w_2 x_2 + w_3 x_3$$

here w_2 can be 0

L1 = feature selection

Here L2 can also make weight 0 right same as L1, ?

Not understood the difference at L1 & L2

L1 = Lasso

L2 = Ridge

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Elastic Net Regularization
combined at L1 & L2

$$\text{Loss} = \text{MSE} + \lambda_1 \sum |w| + \lambda_2 \sum w^2$$

Feature select

1000 $\rightarrow$ 100 $\Rightarrow$ L1 regularization

100 $\rightarrow$ 100 $\Rightarrow$ L2 regularization

Why? Not understood

Q:- why use L1 for feature select why not L2

A:- L1 make weight 0
L2 does not

L2 can also make weight 0
 $\frac{d}{dw} w^2 = 2w = 0$ at $w=0$.

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$$\rightarrow \frac{dL_2}{dw} = 2w$$

Here L_2 will not make the weight 0 it can make weight 0.0000112 but not 0.

Again why? $\Downarrow$

that's why L_2 will not reduce feature L_3 can reduce feature.